

Math 161

Construction of the rational numbers

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1 Equivalence relations

Let us first recall the definition of an equivalence relation.

Definition 1.1. A relation $\sim$ on a set X is called an *equivalence relation* if the following conditions are satisfied for any $x, y, z \in X$:

- (reflexivity) $x \sim x$;
- (symmetry) $x \sim y \implies y \sim x$;
- (transitivity) $x \sim y$ and $y \sim z \implies x \sim z$.

Here's an example of an equivalence relation which we didn't see in class.

Example 1.2. Let $a, b \in \mathbb{Z}$ and $n \in \mathbb{N}$. We say that a is *congruent to b modulo n* and write $a \equiv b \pmod{n}$ if n divides $b - a$, i.e. $b - a = nd$ for some $d \in \mathbb{Z}$. Let us show that congruence modulo n is an equivalence relation. Reflexivity follows from the fact that $a - a = 0 = 0 \cdot n$ for any $a \in \mathbb{Z}$. For symmetry, note that $b - a = nd$ implies that $a - b = n(-d)$. Finally, if $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then we have $b - a = nd$ and $c - b = ne$ for some $d, e \in \mathbb{Z}$, so

$$c - a = c - b + b - a = nd + ne = n(d + e).$$

Thus $a \equiv c \pmod{n}$, and we have shown that congruence modulo n is transitive.

The other key definition is that of an equivalence relation.

Definition 1.3. Let X be a set and $\sim$ an equivalence relation on X . For any $x \in X$, the associated *equivalence class* is the subset of X defined by

$$[x] := \{y \in X \mid x \sim y\} \subset X.$$

In words, the subset $[x]$ consists of all elements of X which are equivalent to x under $\sim$.

Proposition 1.4. Any element of X belongs to exactly one equivalence class.

Proof. Given $x \in X$, we have $x \in [x]$ because $x \sim x$ by reflexivity. To see that this is the only equivalence class containing x , suppose that $y \in X$ is such that $x \in [y]$. Then we must show that $[x] = [y]$.

To see that $[x] \subset [y]$, suppose that $z \in [x]$, i.e. $x \sim z$. The assumption $x \in [y]$ means that $y \sim x$. Thus, by transitivity we obtain $y \sim z$, which says that $z \in [y]$ as needed.

As for $[y] \subset [x]$, suppose that $w \in [y]$, which means that $y \sim w$. We assumed that $x \in [y]$, i.e. $y \sim x$, which implies $x \sim y$ by symmetry. Now transitivity implies that $x \sim w$, or equivalently $w \in [x]$ as desired. $\square$

One says that $\sim$ *partitions* X into equivalence classes. Note that for $x, y \in X$, we have $[x] = [y]$ if and only if $x \sim y$.

The *quotient set* $X/\sim$ of X with respect to $\sim$ is the set of equivalence classes of $\sim$ in X , i.e.

$$X/\sim := \{[x] \mid x \in X\}.$$

Example 1.5. Let us return to the example of congruence modulo n in $\mathbb{Z}$. The set of equivalence classes $\mathbb{Z}/\sim$ is traditionally called the *integers modulo n* and denoted by

$$\mathbb{Z}/n\mathbb{Z} := \{[a] \mid a \in \mathbb{Z}\}.$$

We claim that any equivalence class has the form $[a]$ for some $0 \leq a < n$, and that moreover these equivalence classes are all distinct. In other words, the map

$$\begin{aligned} \{0, 1, 2, \dots, n-1\} &\longrightarrow \mathbb{Z}/n\mathbb{Z} \\ a &\mapsto [a] \end{aligned}$$

is bijective. To see this, recall that any $a \in \mathbb{Z}$ can be divided by n with remainder: that is, we can write $a = nq + r$ where $q \in \mathbb{Z}$ and $0 \leq r < n$. Thus $a - r = nq$, i.e. $a \equiv r \pmod{n}$, or equivalently $[a] = [r]$. On the other hand, if $[a] = [b]$ where $0 \leq a, b < n$, then $a \equiv b \pmod{n}$ means that $b - a = nd$ for some $d \in \mathbb{Z}$. If $d \neq 0$ then

$$|b - a| = |nd| = n|d| \geq n \cdot 1 = n,$$

which is impossible because $0 \leq a, b < n$ implies $|b - a| < n$. Thus $d = 0$, i.e. $a = b$.

2 Rational numbers

We now recall the construction of the rational numbers given in class. Define a relation $\sim$ on the set

$$\mathbb{Z} \times (\mathbb{Z} \setminus \{0\}) = \{(a, b) \mid a, b \in \mathbb{Z}, b \neq 0\}$$

by the formula

$$(a, b) \sim (c, d) \iff ad = bc.$$

Proposition 2.1. *The relation $\sim$ on $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ is an equivalence relation.*

Proof. To prove reflexivity, note that $(a, b) \sim (a, b)$ for any $a, b \in \mathbb{Z}$ with $b \neq 0$, because $ab = ba$ by commutativity of multiplication in $\mathbb{Z}$.

As for symmetry, suppose that $a, b, c, d \in \mathbb{Z}$, $b, c \neq 0$ are such that $(a, b) \sim (c, d)$, i.e. $ad = bc$. Then we have

$$cb = bc = ad = da,$$

again by commutativity of multiplication in $\mathbb{Z}$, which means that $(c, d) \sim (a, b)$.

Finally, to prove transitivity we assume that $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$ for some $a, b, c, d, e, f \in \mathbb{Z}$ with $b, c, f \neq 0$, which means that $ad = bc$ and $cf = de$. This implies that

$$(af)d = a(fd) = a(df) = (ad)f = (bc)f = b(cf) = b(de) = b(ed) = (be)d,$$

where we used associativity and commutativity of multiplication. Since $d \neq 0$ we can apply the cancellative property of multiplication that we proved in class, which implies that $af = be$. This means that $(a, b) \sim (e, f)$ as desired. □

Thus it makes sense to consider the equivalence classes of $\sim$ in $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$. For any $a, b \in \mathbb{Z}$ with $b \neq 0$, we write

$$\frac{a}{b} := [(a, b)]$$

for the corresponding equivalence class: this is the formal definition of a fraction!

Let us check that the following familiar manipulation with fractions is valid.

Proposition 2.2. *If $a, b, c \in \mathbb{Z}$ with $b, c \neq 0$, then $\frac{ac}{bc} = \frac{a}{b}$*

Proof. This follows immediately from the calculation

$$(ac)b = a(cb) = a(bc) = (bc)a$$

using associativity and commutativity of multiplication in $\mathbb{Z}$. □

Now we can define the set of rational numbers.

Definition 2.3. The *set of rational numbers* is defined by

$$\begin{aligned}\mathbb{Q} &:= (\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})) / \sim \\ &= \{[(a, b)] \mid a, b \in \mathbb{Z}, b \neq 0\} \\ &= \{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\}.\end{aligned}$$

Of course, we are interested in more than just the abstract set of rational numbers: we would also like to add and multiply them, as well as compare them with inequalities. For any $a, b, c, d \in \mathbb{Z}$ with $b, d \neq 0$, we define

$$\frac{a}{b} + \frac{c}{d} := \frac{ad + bc}{bd}$$

and

$$\frac{a}{b} \cdot \frac{c}{d} := \frac{ac}{bd}.$$

As simple and familiar as these formulas are, there is something to check here. Remember that the fraction $\frac{a}{b}$ is actually an equivalence class of ordered pairs, and when we write such a fraction we are choosing a single pair (a, b) to represent it. This means that when we define operations like addition and multiplication on such equivalence classes, we need to make sure that our definition is independent of the particular representatives to which we apply the formula.

Proposition 2.4. *The above formulas for addition and multiplication in $\mathbb{Q}$ are well-defined.*

Proof. Suppose that $a, b, a', b' \in \mathbb{Z}$, $b, b' \neq 0$ satisfy $\frac{a}{b} = \frac{a'}{b'}$, i.e. $ab' = ba'$. Then for any $c, d \in \mathbb{Z}$ with $d \neq 0$, we have

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} = \frac{(ad + bc)b'}{(bd)b'} = \frac{adb' + bcb'}{bdb'} = \frac{ba'd + bcb'}{bdb'} = \frac{b(a'd + cb')}{b(db')} = \frac{a'd + cb'}{db'} = \frac{a'}{b'} + \frac{c}{d},$$

where we have used associativity and commutativity of multiplication in $\mathbb{Z}$, as well as Proposition 2.2 above (we combined multiple steps and omitted some parentheses to make this calculation less tedious).

Similarly, one shows that different representatives for the second summand yield the same result. The proof that multiplication is well-defined is completely analogous and left to the interested reader. □

As for inequalities, for any $a, b \in \mathbb{Z}$, $b \neq 0$ we define $\frac{a}{b} > 0$ to mean that either $a, b > 0$ or $a, b < 0$ (i.e. a, b are either both positive or both negative). Again, one has to make sure this is well-defined.

Proposition 2.5. *For any element of $\mathbb{Q}$, the property of being positive does not depend on the choice of fraction used to represent it.*

Proof. Let $a, b \in \mathbb{Z}$, $b \neq 0$ satisfy $\frac{a}{b} > 0$, i.e. $a, b > 0$ or $a, b < 0$. Assume also that $a', b' \in \mathbb{Z}$, $b' \neq 0$ are such that $\frac{a}{b} = \frac{a'}{b'}$, which means that $ab' = ba'$. If $a' = 0$, then we have $ab' = ba' = b \cdot 0 = 0$, and since $b' \neq 0$ this means that $a = 0$, which by trichotomy contradicts our hypothesis that either $a > 0$ or $a < 0$. Thus $a' \neq 0$, so again by trichotomy we have $a' > 0$ or $a' < 0$.

Suppose first that $a' > 0$. If $b > 0$, then $a > 0$ as well, and since

$$ab' = ba' > 0,$$

it follows that $b' > 0$ (since otherwise we would have $ab' < 0$). Similarly, if $b < 0$, then $a < 0$, and then

$$ab' = ba' < 0$$

implies that $b' > 0$. A very similar argument shows that if $a' < 0$, then $b' < 0$ as needed (the interested reader can fill in the details).

□

Finally, one must show that the operations of addition and multiplication on $\mathbb{Q}$, as well as the relation $>$, satisfy Spivak's properties P1-P12. For example, the multiplicative identity element is $\frac{1}{1}$. You'll check that addition is associative on the problem set. The remaining properties P1-P6 and P8-P12 are inherited from the integers in a straightforward way.

Crucially, unlike the integers, the rational numbers have Spivak's property P7 (multiplicative inverses). Namely, if $\frac{a}{b} \neq 0$, which means that $a \neq 0$, then

$$\frac{a}{b} \cdot \frac{b}{a} = \frac{ab}{ab} = \frac{1}{1}$$

by Proposition 2.2. So $\frac{b}{a} = (\frac{a}{b})^{-1}$.